

First name: Alan

Last name: Hossein pour moghadam

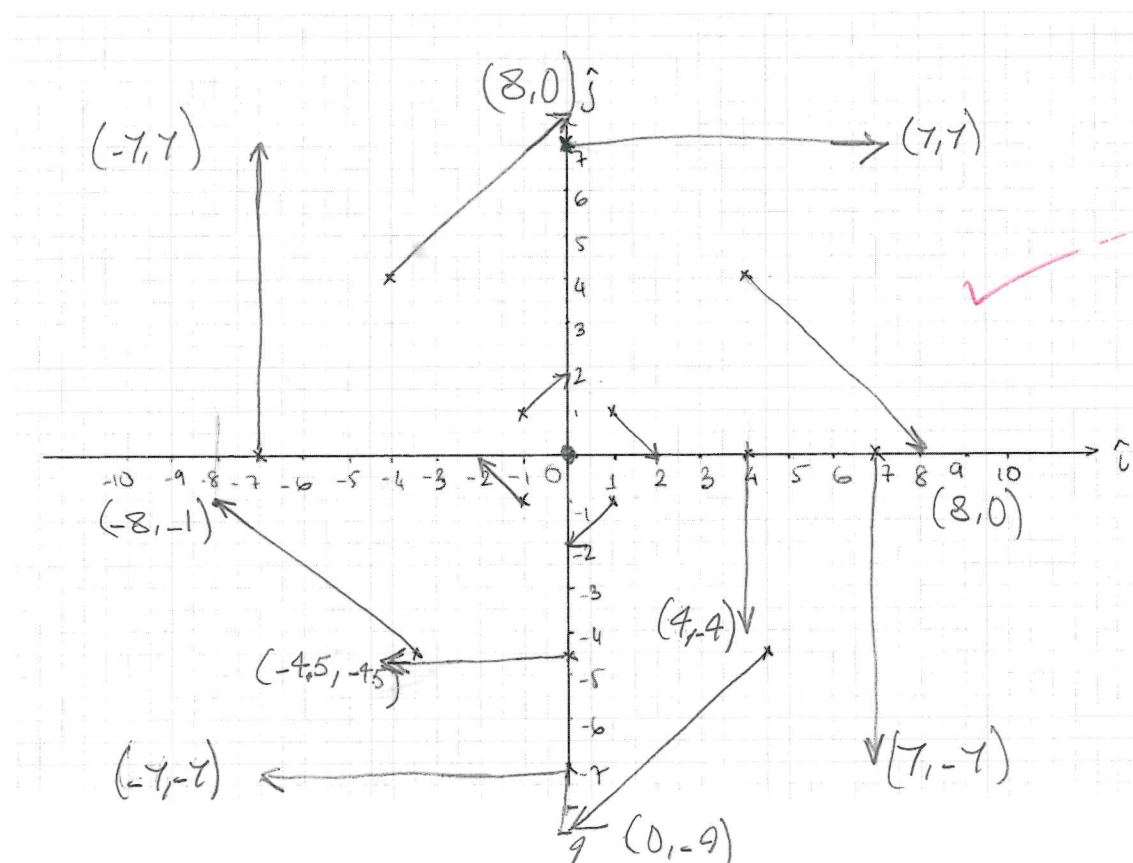
Student ID: 8825132

Problem 1. Sketch vector arrows at the points marked with 'x' to depict the vector field

$$\vec{v} = -y\hat{i} + x\hat{j}.$$

[5 marks]

Solution.



it is a circle with direction of clockwise rotation

□

First name: Alan

Last name: Hossainpour moghadam

Student ID: 8825132

Problem 2. Find the first and second derivatives (with respect to t) of

$$\vec{r}(t) = \begin{bmatrix} 3 \cos(2t) \\ 3 \sin(2t) \\ 4t \end{bmatrix}.$$

[5 marks]

Solution.

First Derivative

$$\vec{r}'(t) = \begin{bmatrix} (2)(-3 \sin(2t)) \\ (2)(3 \cos(2t)) \\ 4 \end{bmatrix} \Rightarrow \begin{bmatrix} -6 \sin(2t) \\ 6 \cos(2t) \\ 4 \end{bmatrix}$$

Second

Derivative :

$$\vec{r}''(t) = \begin{bmatrix} 2(-6 \cos(2t)) \\ 2(-6 \sin(2t)) \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} -12 \cos(2t) \\ -12 \sin(2t) \\ 0 \end{bmatrix}$$

□

First name: *Stuart*Last name: *Murray*Student ID: *8864228*

Problem 1. Sketch vector arrows at the points marked with 'x' to depict the vector field

$$\vec{v} = -y\hat{i} + x\hat{j}.$$

$$\vec{V} = -y + x$$

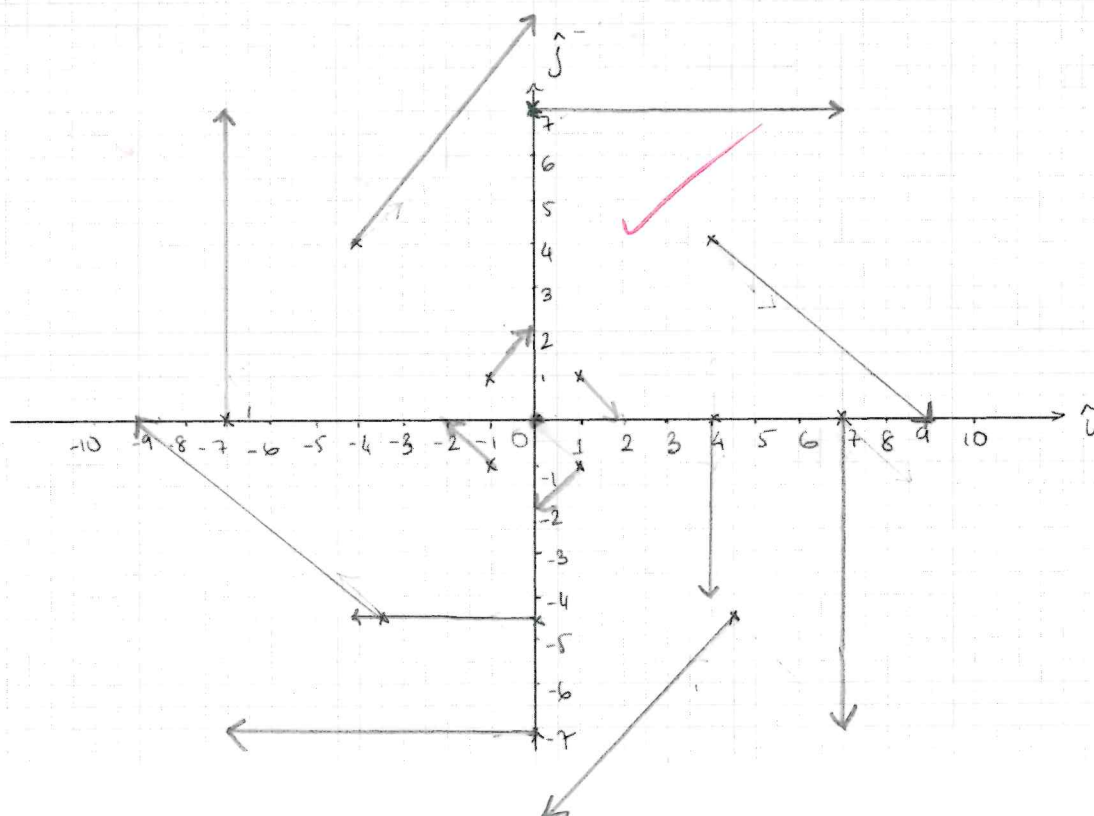
$$\begin{matrix} i = + \\ j = + \end{matrix}$$

$$\vec{V} = -y - x$$

$$\begin{matrix} i = + \\ j = - \end{matrix}$$

[5 marks]

Solution.



First name: *Stuart*Last name: *Murray*Student ID: *8864228***Problem 2.** Find the first and second derivatives (with respect to t) of

$$\vec{r}(t) = \begin{bmatrix} 3 \cos(2t) \\ 3 \sin(2t) \\ 4t \end{bmatrix}.$$

[5 marks]*Solution.*

$$\vec{r}(t) = \begin{bmatrix} 3 \cos(2t) \\ 3 \sin(2t) \\ 4t \end{bmatrix} \rightarrow \vec{r}'(t) = \begin{bmatrix} -6 \sin(2t) \\ 6 \cos(2t) \\ 4 \end{bmatrix} \rightarrow \vec{r}''(t) = \begin{bmatrix} -12 \cos(2t) \\ -12 \sin(2t) \\ 0 \end{bmatrix}$$

$$\frac{d}{dt} \cos(t) = -\sin(t)$$

$$\frac{d}{dt} \sin(t) = \cos(t)$$

□

First name: AMY

Last name: WENTZELL

Student ID: 8820998

Problem 1. Sketch vector arrows at the points marked with 'x' to depict the vector field

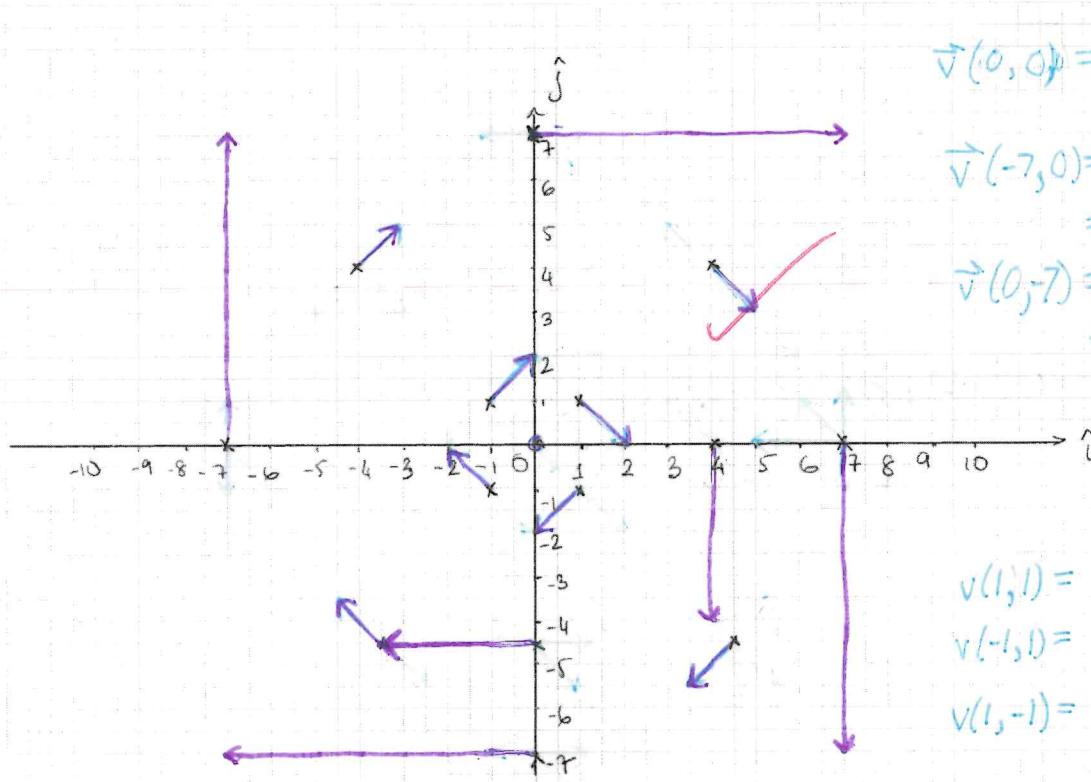
$$\vec{v} = -y\hat{i} + x\hat{j}.$$

[5 marks]

Solution.

$$\vec{v}(0,7) = -y(0) + x(7)$$

$$\vec{v}(7,0) = -y(7) + x(0)$$



$$\vec{v}(0,0) = -y(0) + x(0)$$

$$\vec{v}(-7,0) = -y(-7) + x(0)$$

$$= y(7)$$

$$\vec{v}(0,-7) = -y(0) + x(-7)$$

$$= x(-7)$$

$$v(1,1) = -y + x$$

$$v(-1,1) = y + x$$

$$v(1,-1) = -y - x$$

$$v(-1,-1) = y - x$$

First name: AMY

Last name: WENTZELL

Student ID: 8820998

Problem 2. Find the first and second derivatives (with respect to t) of

$$\vec{r}(t) = \begin{bmatrix} 3 \cos(2t) \\ 3 \sin(2t) \\ 4t \end{bmatrix}.$$

[5 marks]

Solution.

$$\vec{r}'(t) = \begin{bmatrix} -3 \sin(2t)(2) \\ 3 \cos(2t)(2) \\ 4 \end{bmatrix} = \begin{bmatrix} -6 \sin(2t) \\ 6 \cos(2t) \\ 4 \end{bmatrix}$$

$$\vec{r}''(t) = \begin{bmatrix} -(2)6 \cos(2t) \\ -(2)6 \sin(2t) \\ 0 \end{bmatrix} = \begin{bmatrix} -12 \cos(2t) \\ -12 \sin(2t) \\ 0 \end{bmatrix}$$

$$\therefore \vec{r}'(t) = [-6 \sin(2t), 6 \cos(2t), 4]$$

$$\therefore \vec{r}''(t) = [-12 \cos(2t), -12 \sin(2t), 0]$$

□

First name: David

Last name: Toma

Student ID: 8823349

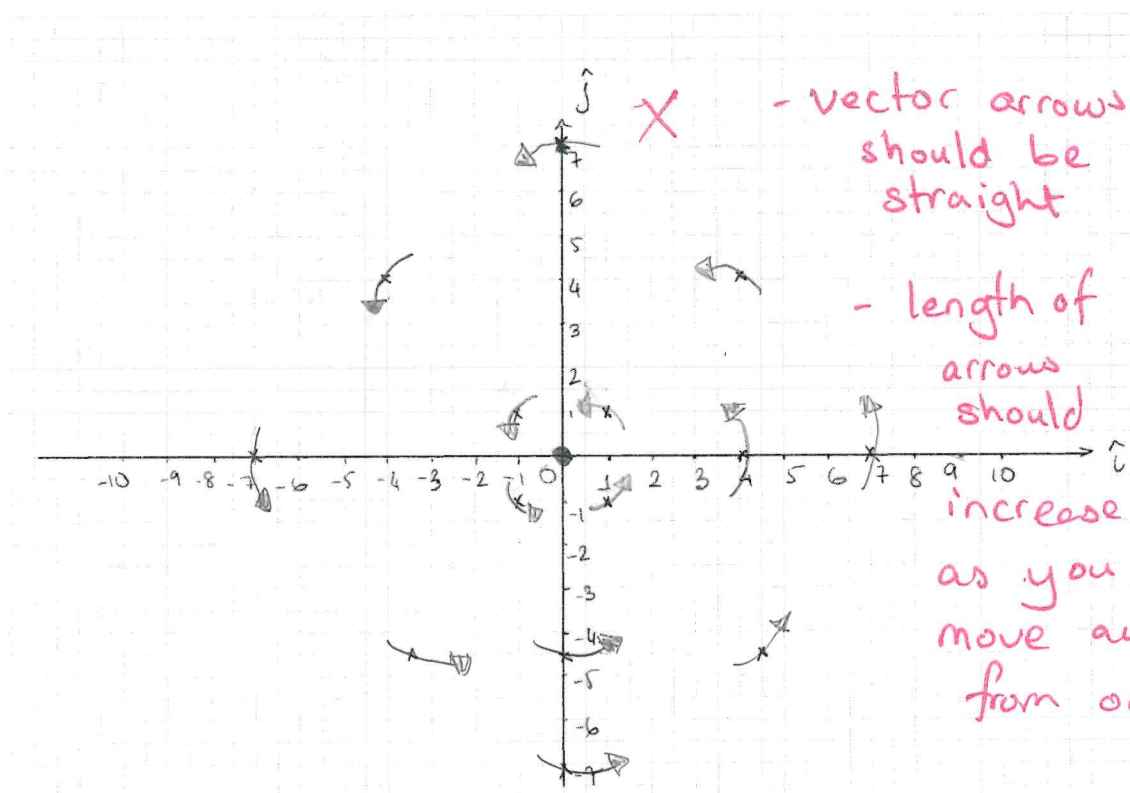
Problem 1. Sketch vector arrows at the points marked with 'x' to depict the vector field

$$\vec{v} = -y\hat{i} + x\hat{j}.$$

1

[5 marks]

Solution.



First name:

Last name:

Student ID:

Problem 2. Find the first and second derivatives (with respect to t) of

$$\vec{r}(t) = \begin{bmatrix} 3 \cos(2t) \\ 3 \sin(2t) \\ 4t \end{bmatrix}.$$

 [5 marks]

Solution.

$$\vec{r}'(t) = \begin{bmatrix} -6 \sin(2t) \\ 6 \cos(2t) \\ 4 \end{bmatrix} \quad \checkmark$$

$$\vec{r}''(t) = \begin{bmatrix} -12 \cos(2t) \\ -12 \sin(2t) \\ 0 \end{bmatrix} \quad \checkmark$$

□

First name: Coulter

Last name: Dolph

Student ID: 8751065

Problem 1. Sketch vector arrows at the points marked with 'x' to depict the vector field

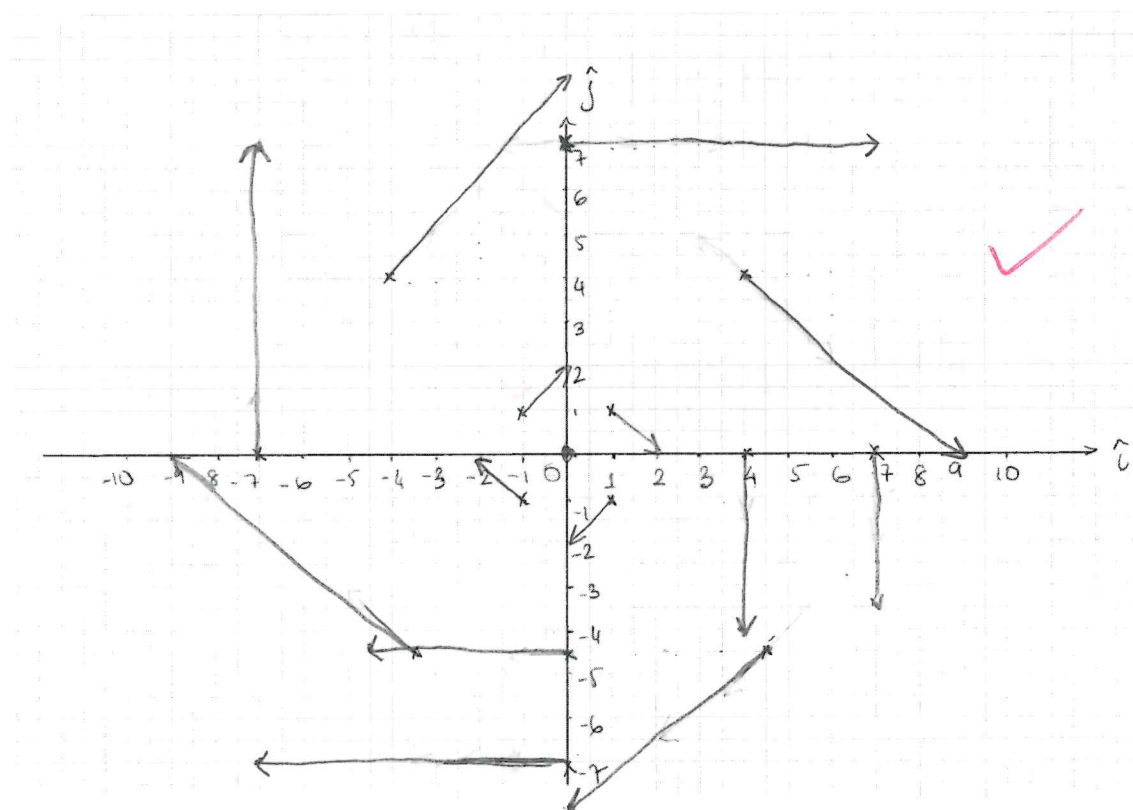
$$\vec{v} = -y\hat{i} + x\hat{j}.$$

[5 marks]

Solution.

$$(1,1) \rightarrow \vec{v} = -1\hat{i} + 1\hat{j}$$

$$(0,-7) \rightarrow \vec{v} = 0\hat{i} - 7\hat{j}$$



First name: Coulter

Last name: Ralph

Student ID: 875 1065

Problem 2. Find the first and second derivatives (with respect to t) of

$$\vec{r}(t) = \begin{bmatrix} 3 \cos(2t) \\ 3 \sin(2t) \\ 4t \end{bmatrix}.$$

[5 marks]

Solution.

$$\vec{r}'(t) = \begin{bmatrix} -6 \sin(2t) \\ 6 \cos(2t) \\ 4 \end{bmatrix}$$

$$\frac{d}{dt} = \cos(t) \\ = -\sin(t)$$

$$\frac{d}{dt} = \sin(t) \\ = \cos(t)$$

$$\vec{r}''(t) = \begin{bmatrix} -12 \cos(2t) \\ -12 \sin(2t) \\ 0 \end{bmatrix}$$

First name: JOSHUA

Last name: DIPLOMA

Student ID: 7897960

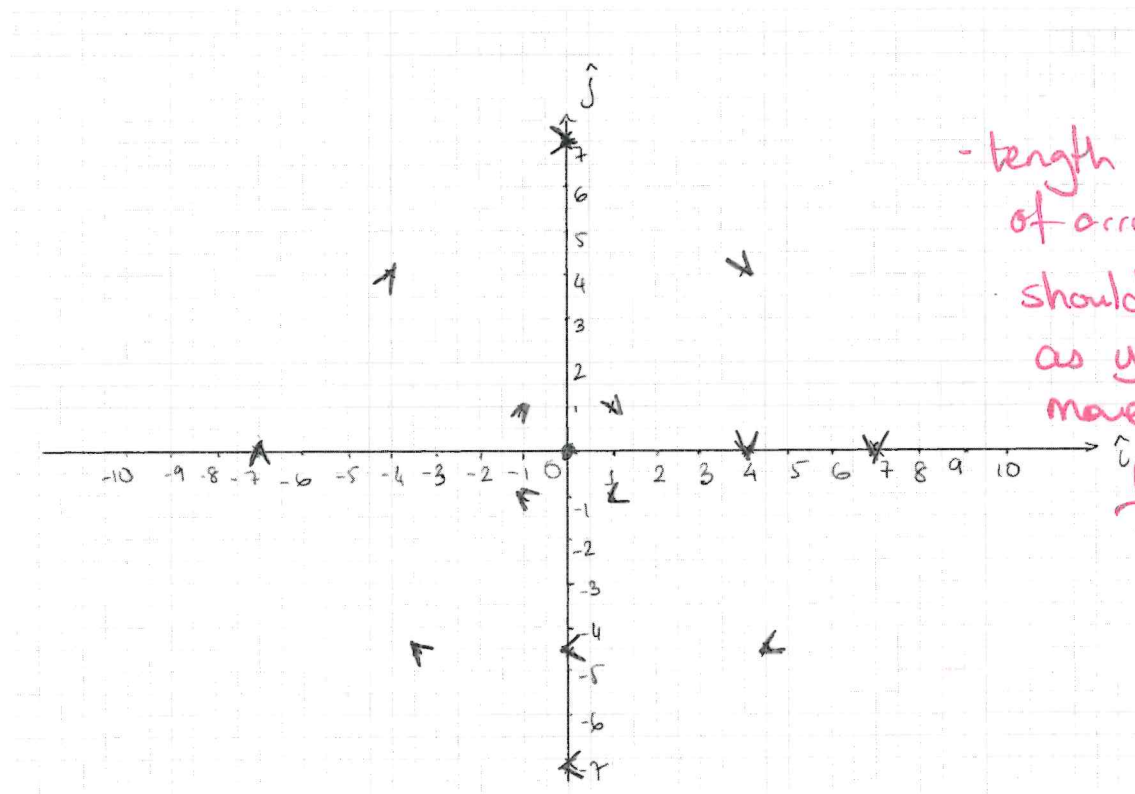
Problem 1. Sketch vector arrows at the points marked with 'x' to depict the vector field

$$\vec{v} = -y\hat{i} + x\hat{j}.$$

3

[5 marks]

Solution.



- length
of arrows
should increase
as you
move away
from (0, 0)

$$\textcircled{1} \vec{v} = \begin{pmatrix} -y \\ x \end{pmatrix}$$

□

First name: JOSHUA

Last name: DIPLOMA

Student ID: 7897960

Problem 2. Find the first and second derivatives (with respect to t) of



$$\vec{r}(t) = \begin{bmatrix} 3 \cos(2t) \\ 3 \sin(2t) \\ 4t \end{bmatrix}.$$

[5 marks]*Solution.*

$$\vec{r}'(t) = \frac{d}{dt} \begin{bmatrix} 3 \cos 2t \\ 3 \sin 2t \\ 4t \end{bmatrix}$$

$$\vec{r}'' = \frac{d}{dt} \begin{bmatrix} -6 \sin 2t \\ 6 \cos 2t \\ 4 \end{bmatrix}$$

$$= \begin{bmatrix} (2)(-3 \sin 2t) \\ (2)(3 \cos 2t) \\ 4 \end{bmatrix}$$

$$\vec{r}'' = \begin{bmatrix} (2)(-6 \cos 2t) \\ (2)(-6 \sin 2t) \\ 0 \end{bmatrix}$$

$$\vec{r}'(t) = \begin{bmatrix} -6 \sin 2t \\ 6 \cos 2t \\ 4 \end{bmatrix}$$


$$\vec{r}'' = \begin{bmatrix} -12 \cos 2t \\ -12 \sin 2t \\ 0 \end{bmatrix}$$


First name: Kyle

Last name: Dick

Student ID: 7206287

Problem 1. Sketch vector arrows at the points marked with 'x' to depict the vector field

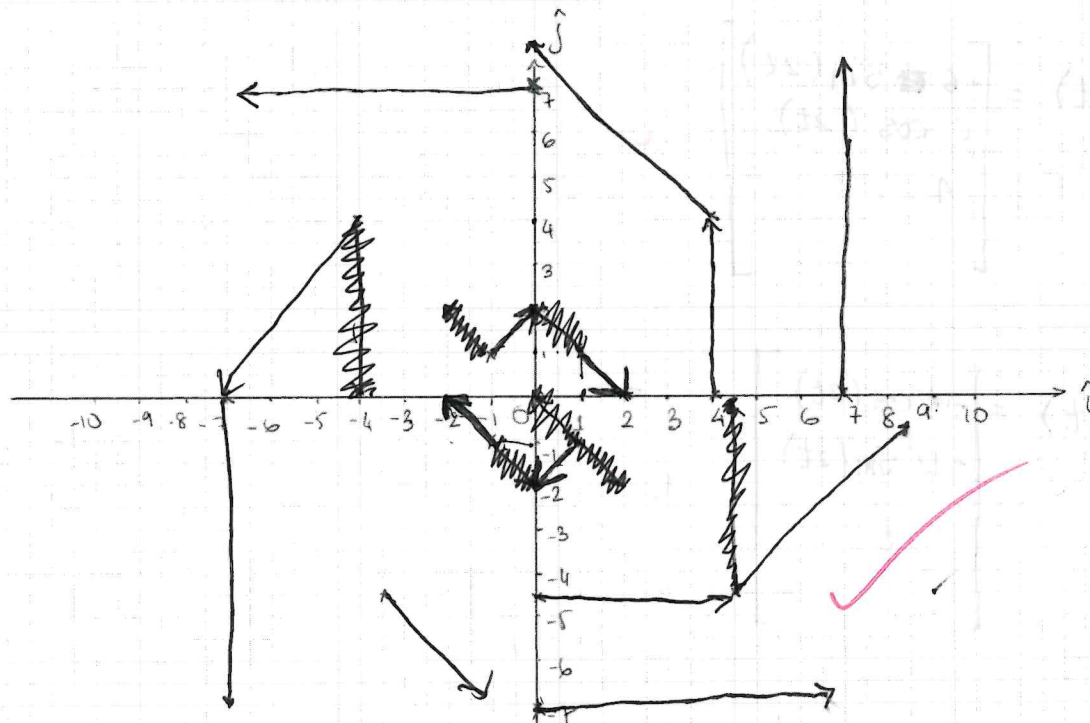
$$\vec{v} = -y\hat{i} + x\hat{j}.$$

[5 marks]

Solution.

$$\begin{aligned} P(1,1): -1\hat{i} + 1\hat{j} & \quad P(-1,-1): 1\hat{i} + -1\hat{j} & P(2,0) = 0\hat{i} + 2\hat{j} \\ P(4,4): -4\hat{i} + 4\hat{j} & \quad P(1,-1): 1\hat{i} + -1\hat{j} & P(-4,0) = 0 + -4\hat{j} \\ P(-1,1): -1\hat{i} + 1\hat{j} & \quad P(4,0): 0\hat{i} + 4\hat{j} & P(-4, -4.5) \end{aligned}$$

4.5 +



$$P(-4, +4)$$

$$+4.5\hat{i} + 4.5\hat{j}$$

First name: Kyle

Last name: Dick

Student ID: 7206287

Problem 2. Find the first and second derivatives (with respect to t) of

$$\vec{r}(t) = \begin{bmatrix} 3 \cos(2t) \\ 3 \sin(2t) \\ 4t \end{bmatrix}.$$

[5 marks]

Solution.

$$\vec{r}'(t) = \begin{bmatrix} -6 \sin(2t) \\ 6 \cos(2t) \\ 4 \end{bmatrix}$$

$$\vec{r}''(t) = \begin{bmatrix} -12 \cos(2t) \\ -12 \sin(2t) \\ 0 \end{bmatrix}$$

□

First name:

Jesse

Last name:

Russell

Student ID:

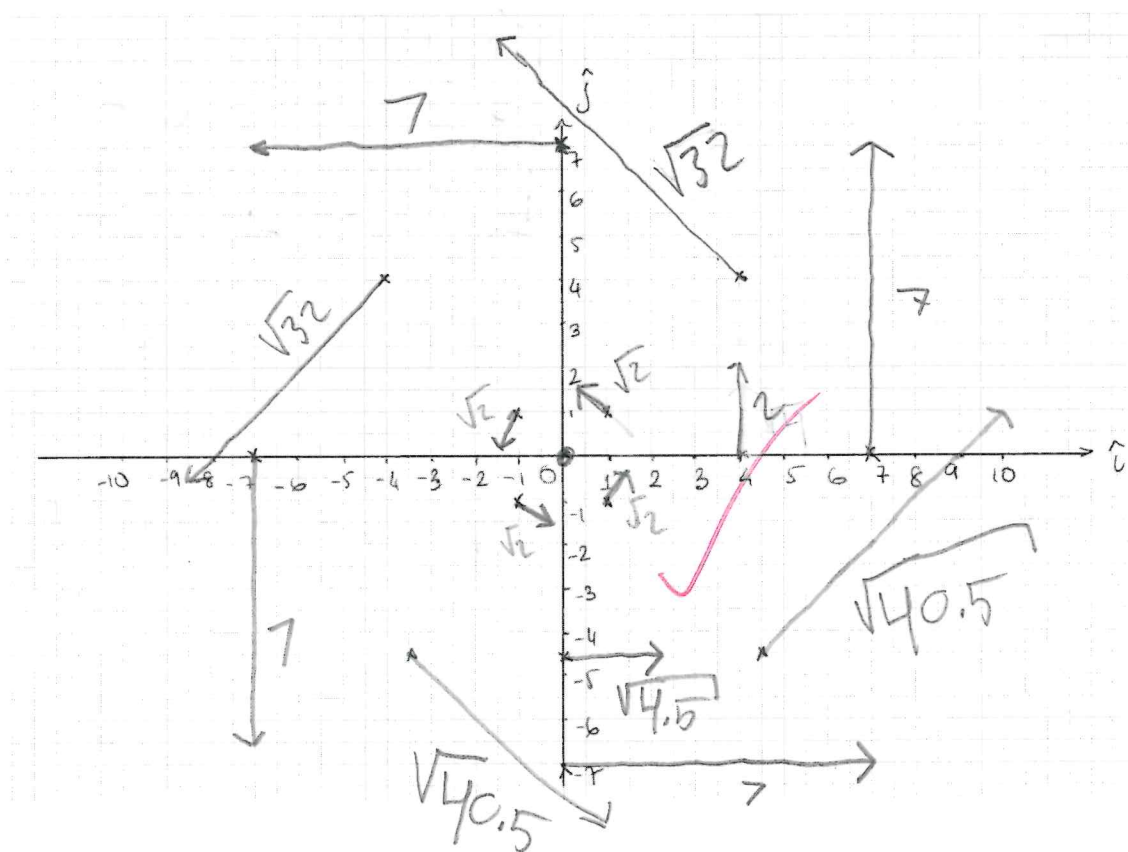
7662240

Problem 1. Sketch vector arrows at the points marked with 'x' to depict the vector field

$$\vec{v} = -y\hat{i} + x\hat{j}.$$

5 [5 marks]

Solution.



First name:

Last name:

Student ID:

Jesse Russell
Russell
7662240

Problem 2. Find the first and second derivatives (with respect to t) of

$$\vec{r}(t) = \begin{bmatrix} 3 \cos(2t) \\ 3 \sin(2t) \\ 4t \end{bmatrix}.$$

[5 marks]

Solution.

$$\vec{r}'(t) = \begin{bmatrix} -6 \sin(2t) \\ 6 \cos(2t) \\ 4 \end{bmatrix}$$

$$\vec{r}''(t) = \begin{bmatrix} -12 \cos(2t) \\ -12 \sin(2t) \\ 0 \end{bmatrix}$$

First name: *Zach*Last name: *Mceman*Student ID: *8768637*

Problem 1. Sketch vector arrows at the points marked with 'x' to depict the vector field

$$\vec{v} = -y\hat{i} + x\hat{j}.$$

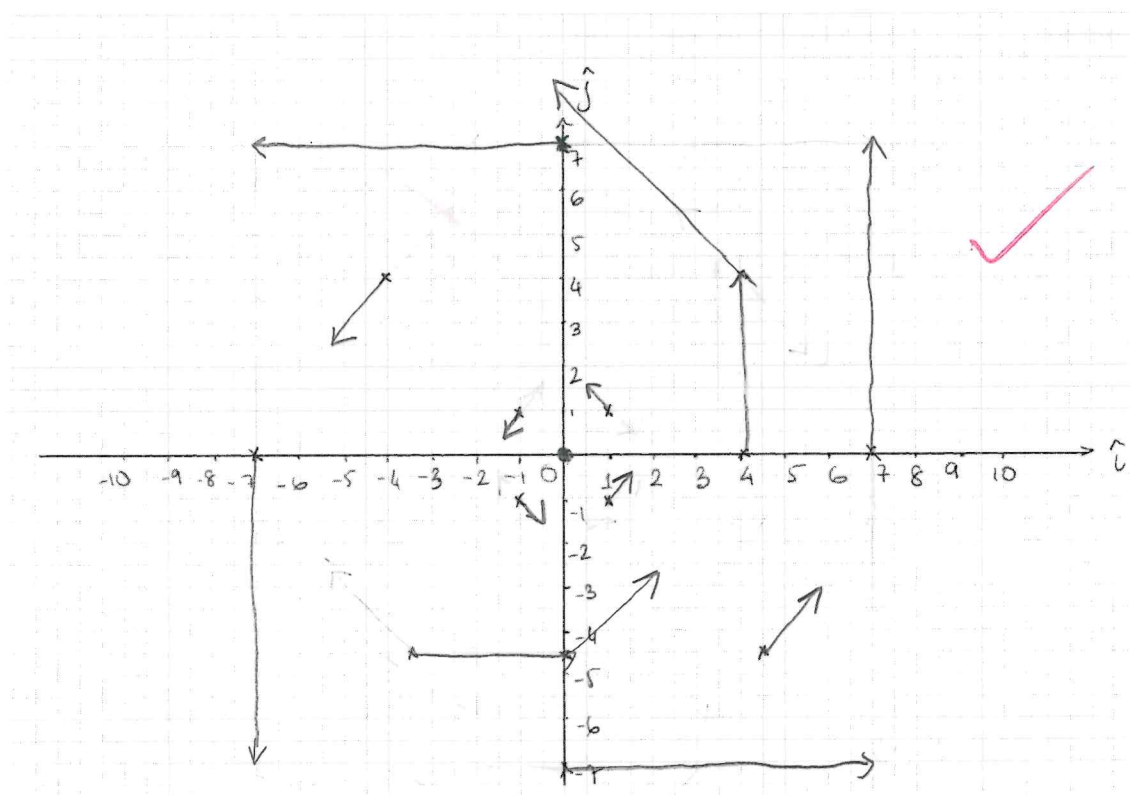
$$\vec{v} = x\hat{j} - y\hat{i}$$

$$(1, -1) \quad (-1, 1)$$

$$(-1, -1) \quad (1, 1)$$

[5 marks]

Solution.



□

First name: Zack

Last name: McEnown

Student ID: 8768637

Problem 2. Find the first and second derivatives (with respect to t) of


$$\vec{r}(t) = \begin{bmatrix} 3 \cos(2t) \\ 3 \sin(2t) \\ 4t \end{bmatrix}.$$

[5 marks]

Solution.

$$\vec{r}'(t) = \begin{bmatrix} -6 \sin(2t) \\ 6 \cos(2t) \\ 4 \end{bmatrix}$$

$$\vec{r}''(t) = \begin{bmatrix} -12 \cos(2t) \\ -12 \sin(2t) \\ 0 \end{bmatrix}$$


First name: Hayden

Last name: Foster

Student ID: 8823130

Problem 1. Sketch vector arrows at the points marked with 'x' to depict the vector field

$$\vec{v} = -y\hat{i} + x\hat{j}.$$

[5 marks]

Solution.

at -4,4

$$\vec{v} = -4\hat{i} - 4\hat{j}$$

at -1,1

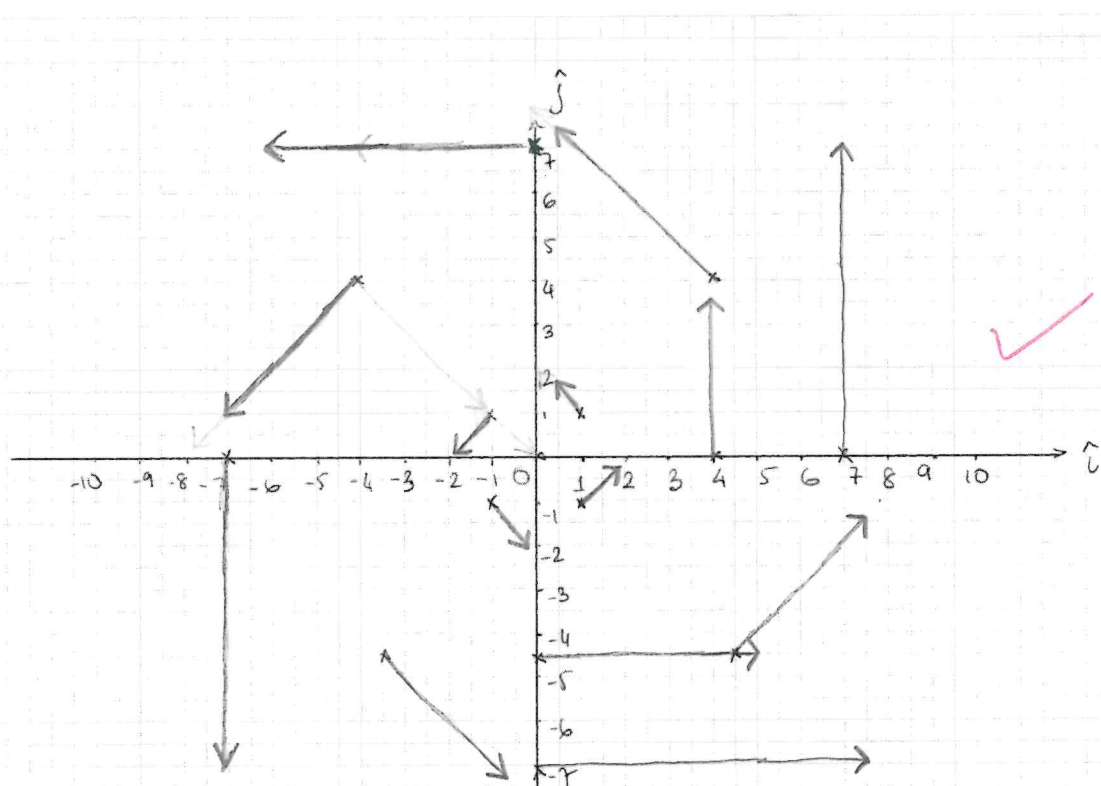
$$\vec{v} = -1\hat{i} - 1\hat{j}$$

at 1,1

$$\vec{v} = -1\hat{i} + 1\hat{j}$$

at 4,4

$$\vec{v} = -4\hat{i} + 4\hat{j}$$



at 4,0

$$\vec{v} = 0 + 4\hat{j}$$

at 0,7

$$\vec{v} = -7\hat{i} + 0$$

at 0,-4.5

$$\vec{v} = +4.5\hat{i} + 0$$

at -1,-1

$$\vec{v} = +1\hat{i} - 1\hat{j}$$

at 1,-1

$$\vec{v} = 1\hat{i} + 1\hat{j}$$

~~at 4,0~~
 ~~$\vec{v} = 0 + 4\hat{j}$~~

at 7,0

$$\vec{v} = 0 + 7\hat{j}$$

at 0,-7

$$\vec{v} = 7\hat{i} + 0$$

□

at -7,0

$$\vec{v} = 0 - 7\hat{j}$$

at -3.5,-4.5

$$\vec{v} = +4.5\hat{i} - 3.5\hat{j}$$

at 4,-4.5

$$\vec{v} = +4.5\hat{i} + 4\hat{j}$$

First name:

Hayden

Last name:

Foster

Student ID:

Problem 2. Find the first and second derivatives (with respect to t) of

$$\vec{r}(t) = \begin{bmatrix} 3 \cos(2t) \\ 3 \sin(2t) \\ 4t \end{bmatrix}$$

[5 marks]

Solution.

$$3 \frac{d}{dt} \cos(2t)$$

$$= -6 \sin(2t)$$

$$3 \frac{d}{dt} \sin(2t)$$

$$= 6 \cos(2t)$$

$$4 \frac{d}{dt} t$$

$$= 4$$

$$\vec{r}' = \begin{bmatrix} -6 \sin(2t) \\ 6 \cos(2t) \\ 4 \end{bmatrix}$$

$$\vec{r}'' = \begin{bmatrix} -12 \cos(2t) \\ -12 \sin(2t) \\ 0 \end{bmatrix}$$

$$-6 \frac{d}{dt} \sin(2t)$$

$$= -12 \cos(2t)$$

$$6 \frac{d}{dt} \cos(2t)$$

$$= -12 \sin(2t)$$

$$\frac{d}{dt} 4$$

$$= 0$$